Minseok Kim

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github.com/pysunn14

Education

Ajou University *Mar 2021 – Feb 2027*
B.S. in Software and Computer Engineering (Advanced Program), expected

- GPA: 4.25 / 4.5, Major GPA: 4.22 / 4.5, Class Rank: 11 / 347

Research Interests

Algorithm, On-Device AI, LLM

Skills

Python, C/C++, Swift, SQL, PyTorch

Experience

iKnow Lab, Ajou University *Jul 2025 – Mar 2026*
Undergraduate Researcher / Advisor: Prof. Hyunsouk Cho

- Observed that 3 of 4 open-source VLMs reason less accurately about vertical spatial relations than horizontal ones, with a gap of **up to 52.12 percentage points**. Hypothesized that because the ViT flattens image patches in row order, an object one row below sits 24 tokens away in the sequence, and RoPE's relative distance decay weakens attention between the two objects
- Designed 2,000 synthetic samples that vary only the 1D patch sequence distance while holding the 2D distance fixed, and analyzed attention in LLaVA-Next 7B. Found a negative correlation between 1D patch sequence distance and inter-object attention (**Spearman's $\rho = -0.782$**)

AI·SW Maestro, 17th Cohort (Seoul) *Apr 2026 – Present*
Trainee / Completion expected Nov 2026

- Hosted by the Ministry of Science and ICT (MSIT) / Organized by IITP and the Federation of Korean Information Industries (FKII)
- Responsible for SLM optimization and mobile feature development in 'Byeolmuri', a three-person team project. In TestFlight beta and preparing for public release

A.N.S.I. *Mar 2024 – Present*
Ajou University Algorithm Club

- Solved algorithm problems in C++ and ran study sessions for over two years. Joined vacation coding camps and ICPC preliminary team training
- Competed in SCPC, UCPC, and the ICPC Seoul Regional preliminary
- Placed 21st in the 2025 shake! finals as an Ajou University representative (finals held Jan 2026)

Armed Forces Command and Communications Command *Mar 2022 – Sep 2023*
Cyber Network Operations Center (CNOC) / CERT Soldier / Army Sergeant, Service Completed

- Blocked unauthorized devices and handled incidents through NAC monitoring

Publications

Diagnosing Axis-Dependent Spatial Bias in Vision-Language Models *Jun 2026*
First Author / Korea Computer Congress (KCC 2026) Undergraduate Paper Competition / Poster

Projects

Byeolmuri *May 2026 – Present*

Mobile On-Device AI Agent App / AI Development / AI·SW Maestro

github.com/mornye-minor-gallery/PetAI-AI

- Owned Gemma 4 E2B on-device inference for an agent app where users talk with a character that runs device functions. Bridged Unity and Swift through a C ABI, ported the Swift core to Android with the Swift SDK for Android, and built long-term memory with EmbeddingGemma and SQLite
- Traced the first-response delay reported by beta users to recomputing the fixed prompt on every request, and applied KV cache reuse. Cut mean TTFT on iPhone 17 CPU from 4.920 s to 1.274 s, a **74.1% reduction**, with unchanged outputs
- Forked the LiteRT-LM C++ engine to add a state transfer API that keeps the cache across restarts
- Fixed the XNNPACK buffer calculation behind a memory spike at longer context lengths, found through allocation tracing. Cut peak iPhone memory by **about 0.8 GiB** with unchanged outputs
- Applied an MLP intent classifier on EmbeddingGemma embeddings to stop ordinary conversation from triggering tool execution, **reducing false positives from 19.3% to 2.9%** on a new 280-item evaluation set. Tool arguments are validated in Swift before execution
- Added query selection by utterance type (Topical/NonTopical) for few-shot example retrieval that did not fit the conversation, **raising accuracy from 50% to 91.9%** on the test set

TOWA *Mar 2026 – Jun 2026*

AI Manga Translation Workstation / AI Development in a Team Project

github.com/trit-ajou/TOWA

- Designed and built the Model Engine connecting CRAFT text detection, OCR, LLM translation, and inpainting. Since the final image alone did not show where quality dropped, restructured it to store each stage's inputs, outputs, and artifacts per transaction
- Generative inpainting redrew artwork outside the text and left original characters behind, so composited its output only inside the text mask and removed the remaining text with OpenCV Telea. Reduced the share of pixels changed from the original **from 23.15% to 1.50%** and kept everything outside the mask untouched
- Cut translation, inpainting, and typesetting time by **about 60%** across three real manga. Demonstrated at the AJOU SOFTCON booth

Honors and Scholarships

Third Prize (5th Overall), 5th Inter-University Deep Learning Challenge *Oct 2026*
SW-Centered University Programs of Kyungpook, Kyung Hee, Pusan, Ajou, Chonnam, and Jeonbuk

Honorable Mention, 25th TOPCIT (Ajou University Top Scorers) *May 2026*
Director's Award, Ajou University AI Convergence Education Institute

Excellence Award, Ajou Programming Contest (APC) Div. 1 *May 2026*
Ajou University SW-Centered University Program

Second Prize (3rd Overall), 4th Inter-University Deep Learning Challenge *Oct 2025*
SW-Centered University Programs of Kyungpook, Kyung Hee, Pusan, Ajou, Chonnam, and Jeonbuk

6th Overall / Ajou First-Place Special Award, 2025 Centroid Cup *Sep 2025*
Inter-university algorithm contest of Kookmin, Chung-Ang, Inha, and Ajou Universities

Excellence Award, Ajou Hackathon *May 2025*
Ajou University SW Convergence Education Institute

Two Excellence Awards and One Encouragement Award, Mogakso *2024 – 2026*
Excellence: Winter 2024, Summer 2025 / Encouragement: Summer 2026

Yulgok Academic Merit Scholarship, Four Semesters *2021 – 2026*
Spring 2021, Spring 2025, Fall 2025, Spring 2026